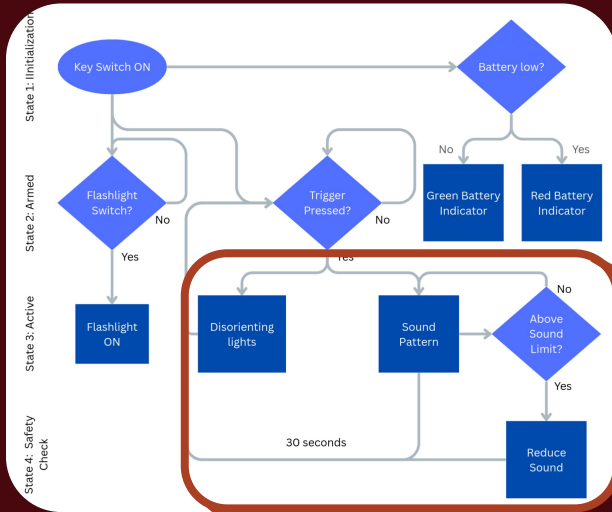


Initial Design Review

Christina Walker, Diego Cummins, Jessy Pena, Jose Aguirre, Noah
Schatz

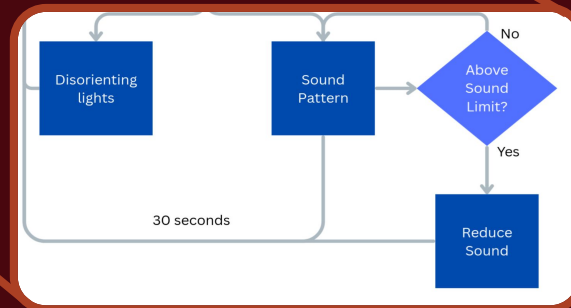


Christina Walker

Data Acquisition

Primary Responsibility: Developing data acquisition and logging system

- Role in the project:
 - Data logging system for speaker, microphone, and LEDs
 - Monitoring of sound and light outputs
 - System safety, testing, and validation
- Contributions I've made
 - Data acquisition for microphone input
 - Real-time sound level logging system
 - Tested speaker and LEDs



Here's what I did:

	COMPLETED	IN PROGRESS	NOT STARTED
Data logging system for speaker, microphone, and LED outputs	-	50%	-
Integrate sensor data with microcontroller	-	45%	-
Basic data storage/logging functionality		-	-
Support system testing and validation	-	30%	-

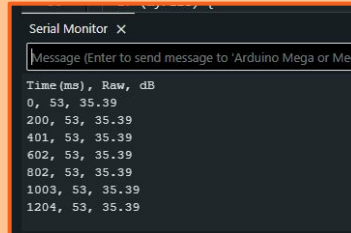
Issues I had:

1. Problems

- Design changes
- Uncertainty in final system parameters
- Limited progress on implementation
- Need to ensure compatibility between multiple subsystems

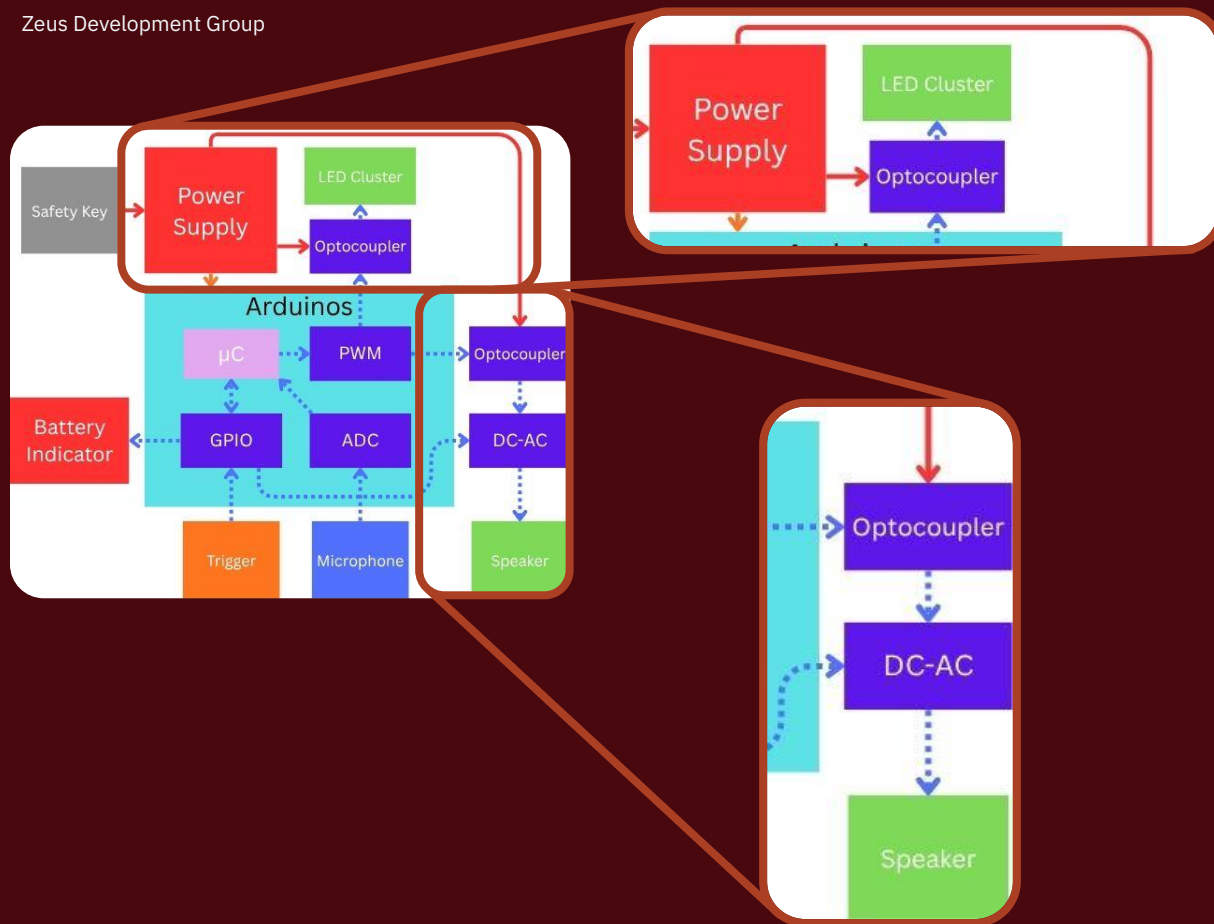
2. Solutions

- Design a flexible logging system that can adapt to changing inputs
- Use placeholder values for testing until specifications are finalized



3. Projected Impact

- Minor delays expected due to changes in requirements
- Estimated completion of data logging implementation: within 1 week
- Estimated completion of integration and testing: within 2–3 weeks



Diego Cummins

Power Systems Engineer

- Project Role
 - Design and analyze the power architecture for Zeus.
 - Design power converters for the LED and speaker subsystems
 - Define power requirements, peak and average power budgets, and duty-cycle limitations to ensure safe
- Summary of my contributions
 - Created a model of the housing for the lights.
 - Done simple calculations for the power requirements of the lights.
 - Creating model of exterior of the device.

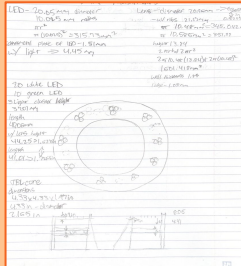
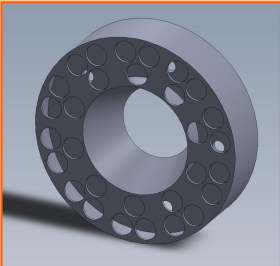
Here's what I did:

	COMPLETED	IN-PROGRESS	NOT STARTED
Calculated power requirement for LEDs		100%	-
LED housing model		100%	-
Model of exterior of device	-	30%	-
Design of power converters	-	-	
Create DC to AC Converter Circuit for speaker	-	-	

Issues I had:

1. Problems

- Minimal space to work with while modeling the LED housing.
- Sudden changes in design
- Parts not meeting what they advertised to output.
- Understanding of power converters



2. Solutions

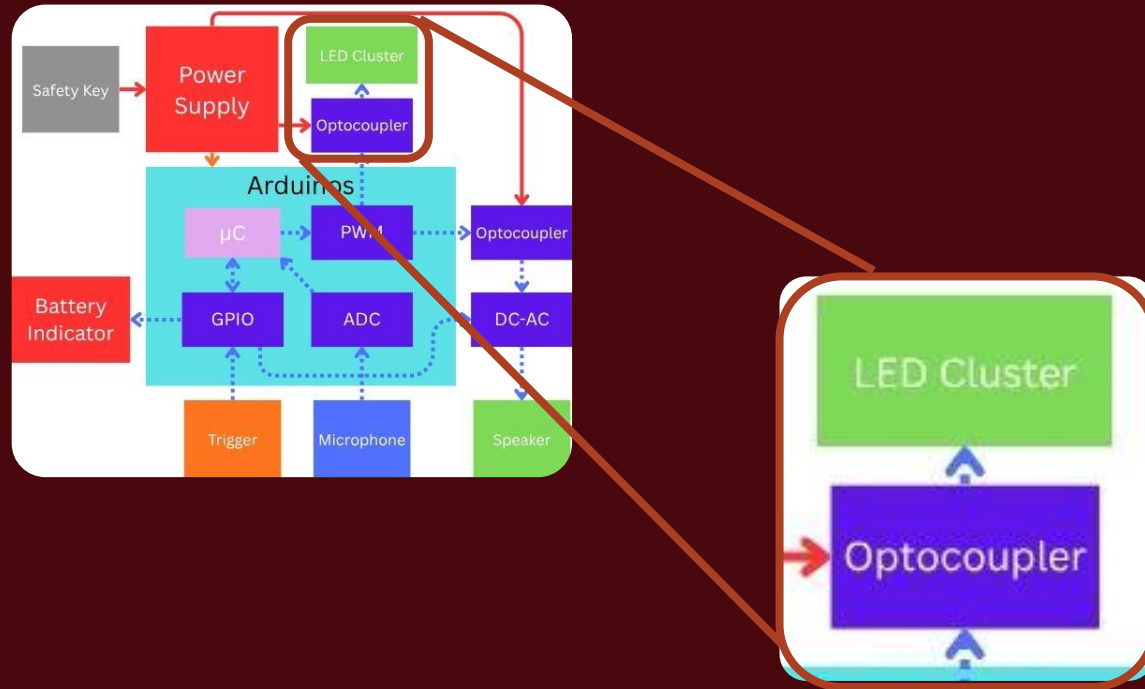
- Pivot into using less LEDs
- Redesign of entire device caused changes in measurements.
- Overcompensation of power requirements in order to handle added strain if needed.
- Researched converters and have done simulations in other power focused classes.

3. Projected Impact

- Improper heat sinking
- Increase in modeling time investment
- Delay in converter design as well as errors in calculations and efficiency.

Jessy Pena

Hardware Integration Engineer



- My role is to:
 - Verify compatibility of various components
 - Design interior framework of device
 - Compile circuit into comprehensive PCB
- A summary of my contributions:
 - Soldered & tested LEDs' working and max loads
 - Built circuit on breadboard to verify correct power distribution
 - Creating CAD model of interior components

Here's what I did:

	COMPLETED	IN PROGRESS	NOT STARTED
Soldered & Tested LEDs Max & Working Loads		100%	-
Verify compatibility of components	-	50%	-
Create CAD Model of interior	-	20%	-
Compile circuit into comprehensive PCB	-	0%	

Issues I had:

1. Problems

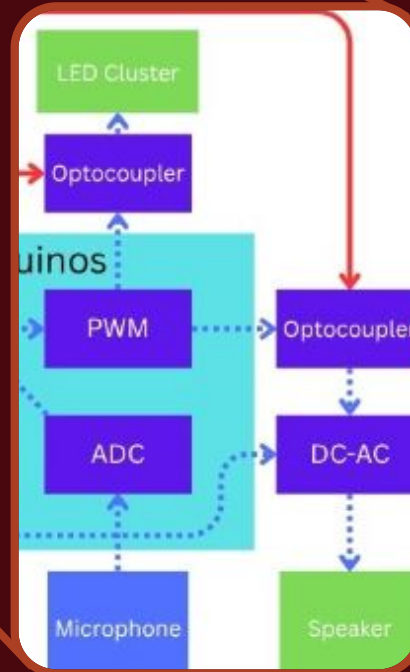
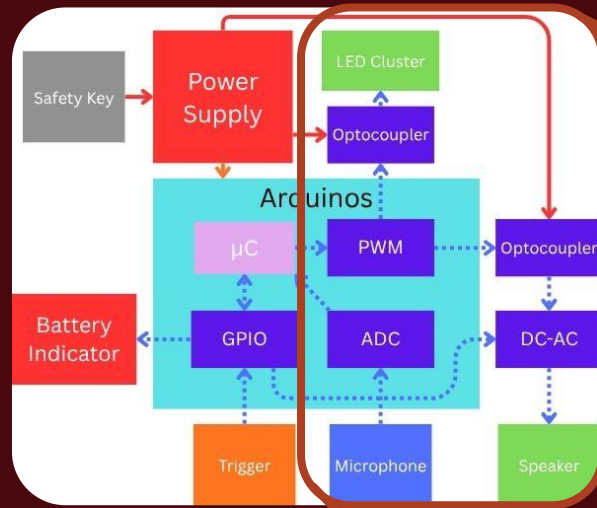
- Indecision on what components would be best to use for the device
- Cross-platform incompatibility between Solidworks & Fusion360

2. Solutions

- Clear communications of updates on findings with safety requirements research
- Need to learn SolidWorks in order to properly integrate interior with OR keep files separate

3. Projected Impact

- Delay created ordering components, therefore testing may be delayed by potentially weeks
- Delay with learning software OR potential incompatibility issues when printing parts



Jose Aguirre Embedded Systems Engineer

- Designed circuits to be implemented in the device.
 - Researched on LED's with the needed lumen.
 - Designed LED circuit to implement into the microcontroller.
 - Tested LED's actual limits in order to have more accurate data.
- This is a summary of the biggest contributions I've made.
 - Designed the Draft LED circuit.
 - Tested LED's limit.

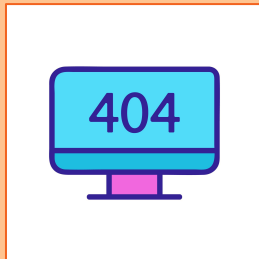
Here's what I did:

	COMPLETED	IN-PROGRESS	NOT STARTED
LED Circuit Design		-	-
LED Circuit Simulation	-	90%	-
Speaker Circuit Design		-	-
Speaker Circuit Simulation	-	-	
LED and Speaker Circuit Design	-	-	

Issues I had:

1. Problems

- Complex LED's are needed for the desired intensity.
- Circuit Design programs complexity needed to simulate specific components properly.



2. Solutions

- Use resistors for each LED in order to control every individual LED thru the microcontroller.
- Learn PSpice to a more complex level than what I have used, or change programs.

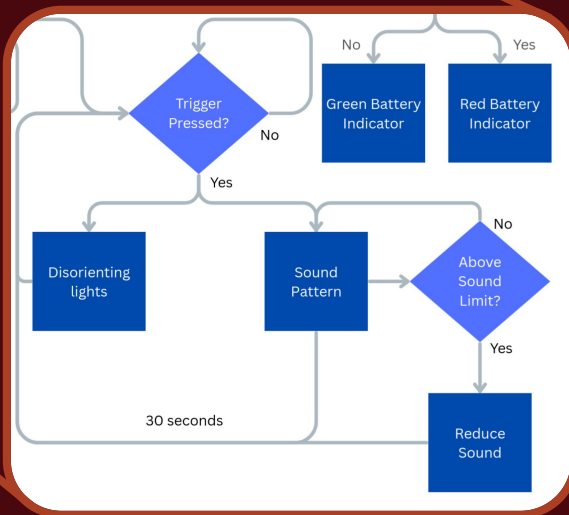
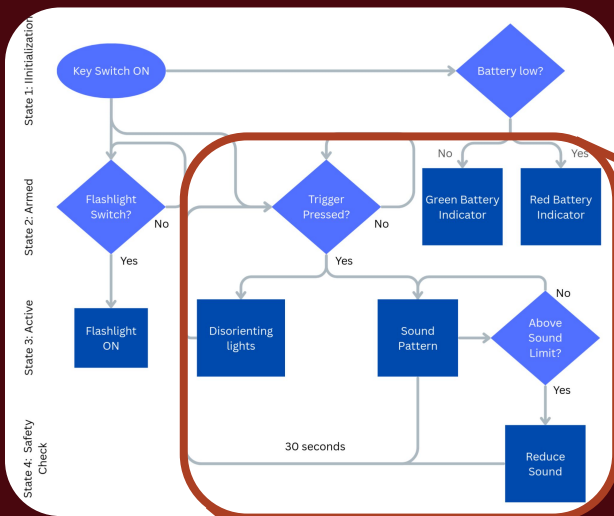
3. Projected Impact

- Can't properly test LED circuit before implementation without damaging the components.

Noah Schatz

Control Systems and UI Engineer

- Designed algorithms to regulate the sound and light patterns
 - Researched light strobe frequency
 - Experimented with different low freq patterns
 - Tested various sound wave shapes and frequencies
- This is a summary of the biggest contributions I've made.
 - Determined ideal sound frequencies
 - Determined most disorienting light flicker and strobe



Here's what I did:

	COMPLETED	IN-PROGRESS	NOT STARTED
Determined most effective light flickering frequency		-	-
Programmed various “random” strobe patterns to cycle through	-	90% *	-
Programmed green light cycle.		-	-
Determined ideal sound frequencies		-	-
Programming the sound	-	-	

* Needs further testing with a completed prototype

Issues I had:

1. Problems

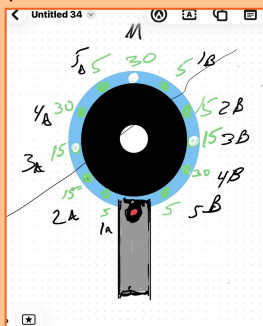
- No major technical errors
- Changing requirements for the LED and sound algorithms
 - Different number of lights
 - Different light functions and patterns

2. Solutions

- Discussed what areas needed to be backed by our own development (for the sake of the course)
- Set a framework of communication in terms of how adjustments and alterations would be made

3. Projected Impact

- Software development should progress fairly quickly now that the number of changes has stabilized.
- The majority of the work will be in testing and refinement
 - The remaining code shouldn't be difficult to draft



Thank you!

Any questions?

Contact
zeusdevgroup@gmail.com